

Arrowized Functional Reactive Programming

Signal functions for game loops

Functional Reactive Programming (FRP) λ

Model values which change over time

Classical FRP

Monadic

Arrowized FRP

Arrowized

Functional Reactive Programming (FRP) λ

Model values which change over time

Classical FRP

Monadic

Time $\rightarrow a$

10

50

30

Time

t_0

t_1

t_2

Arrowized FRP

Arrowized

Time $\rightarrow a \rightarrow b$

$\lambda a \rightarrow a + 10$

$\lambda a \rightarrow a \times 22$

$\lambda a \rightarrow a - 10$

Signals

λ

`data Signal a b`

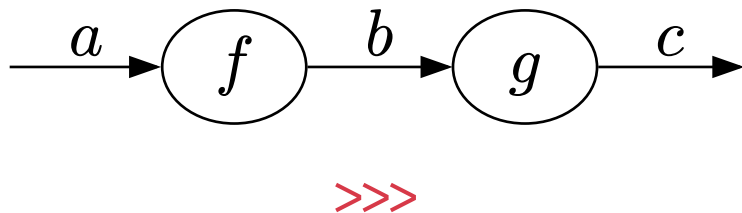
Like a function from a to b .

Signals

λ

`data Signal a b` Like a function from a to b .

`(>>>) :: Signal a b → Signal b c → Signal a c`



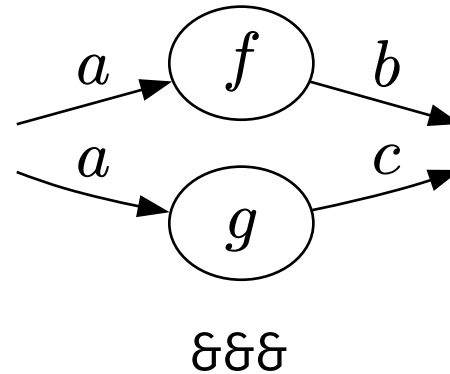
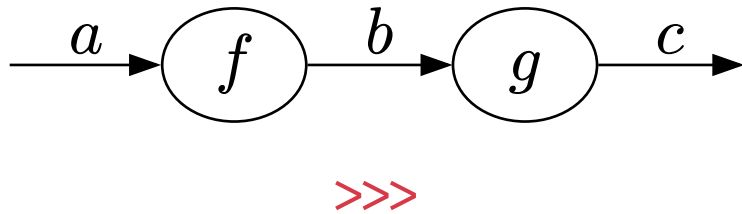
Signals

 λ

`data Signal a b` Like a function from a to b .

`(>>>) :: Signal a b → Signal b c → Signal a c`

`(&&&) :: Signal a b → Signal a c → Signal a (b,c)`

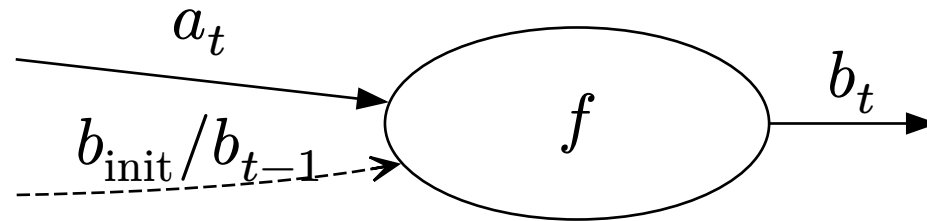


for more details, see Yampa package and <https://www.haskell.org/arrows>

Stateful Signals

λ

`feedback` :: $b \rightarrow \text{Signal } (a, b) \rightarrow \text{Signal } a \rightarrow b$



feedback

Simulation with Signals

λ

Signals match well with simulation / game loops

Simulation with Signals

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```
myCounter = 0
myGameState = ...

while (!end) {
    myCounter = myCounter + 10
    myGameState = ...
}
```

Switching

λ

Compared to monads, arrows have static control flow

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Compared to monads, arrows have static control flow

Switching allows for dynamic control flow when needed

```
switch :: Signal a (b, Maybe c) →  
  (c → Signal a b) → Signal a b
```

Switching

λ

Compared to monads, arrows have static control flow

Switching allows for dynamic control flow when needed

```
switch :: Signal a (b, Maybe c) →  
  (c → Signal a b) → Signal a b
```

```
rSwitch :: Signal a (b, Maybe c) →  
  (c → Signal a (b, Maybe c)) → Signal a b
```

More things

λ

- Simple implementation of signal functions
- Reactimate vs Yampa vs Dunai
- AFRP and classical FRP
- AFRP vs ECS for game development
- Multithreading & Synchronizing different sampling rates